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Data Science FINAL PROJECT REPORT

Project Title:

Predicting Real Estate Listing Success and Price Optimization Using Structural, Locational, and Textual Market Features

Student Name and SRN:

Reddy Abinaya

7235SYXD

Supervisor: Tai Hock Lin, Andy.

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GitHub address: https://github.com/7235SYXD/Real-Estate.git.

DECLARATION STATEMENT

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Student SRN number: 7235SYXD

Student Name printed: Reddy Abinaya

Student signature: Abinaya

UNIVERSITY OF HERTFORDSHIRE

SCHOOL OF PHYSICS, ENGINEERING AND COMPUTER SCIENCE

Predicting Real Estate Listing Success and Price Optimization Using Structural, Locational, and Textual Market Features

Abstract

In this report, we describe an end-to-end data science workflow for predicting listing prices and the outcomes in Texas and New York residential markets. We utilized four public Kaggle datasets which comprises of over 2.8 million rows, which were cleaned and joined to form a dataset augmented with the eight artificial variables and a natural language processing profile for each state based on ~21K property listings analyzed with the VADER sentiment analyzer. These modelling corpus consisted of 51 variables and ~120K row listings selected by stratified sampling and was used for training, hyperparameter tuning with Optuna and Bayesian optimization, and stacking of four base models (Random Forest, CatBoost, a residual multilayer perceptron, and HistGradientBoosting), which produces a Ridge-stacked ensemble with test-set R^2 of 0.7869 and real-dollar mean absolute error of $62,473. The ablation study conducted with the three reduced feature sets showed that location encodings account for RMSE decrease of 0.3073 compared to 0.0013 attributable to the NLP profile, thus demonstrating the superiority of location-encoded variables in price prediction with textual variables aggregated at the state level. Interpretability with the SHAP showed 70.2%, 29.6%, and 0.2% model importance attributed to structural, location, and NLP features correspondingly.

The secondary classification of the binary task, which is predicting the listing success, yielded the highest test, where AUC-ROC was 0.7707 through HistGradientBoosting. The whole process was conducted on seven linked Google Colab notebooks.

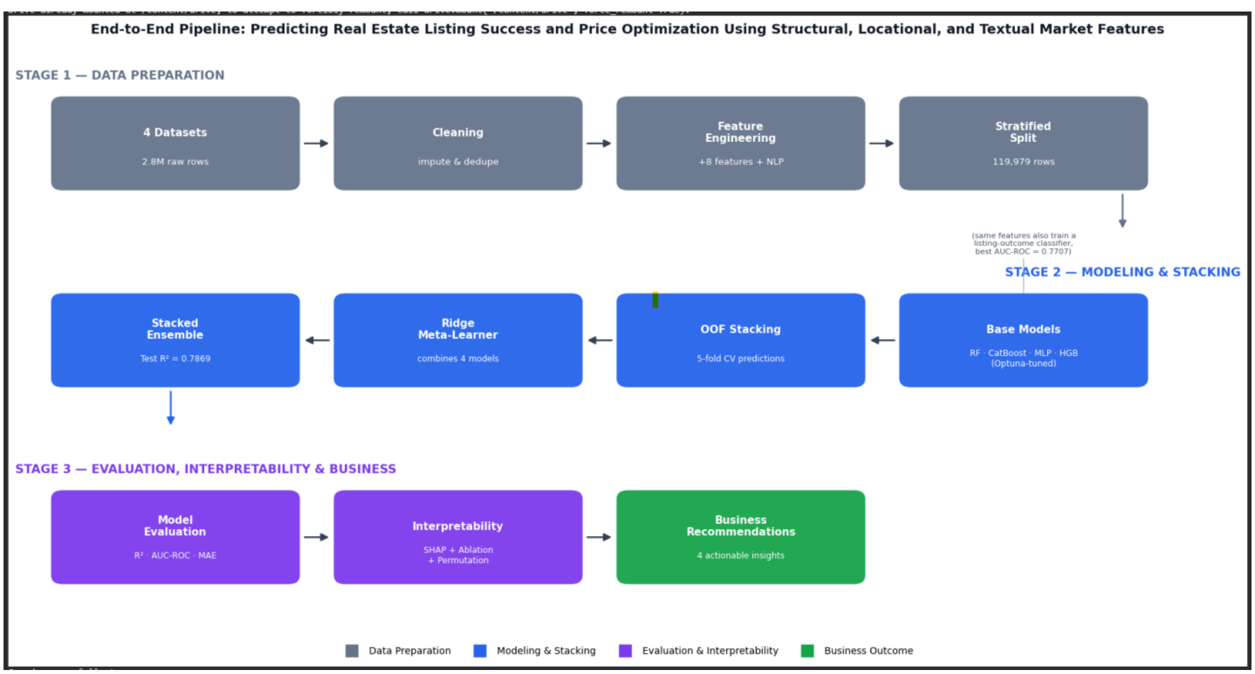

FIG 1: END TO END PIPELINE

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1. Introduction

1.1 Background and Motivation

- Correctly valuing the residential real estate is an interdisciplinary task where 3 main important factors meet, the first one is economics, second one is computer science, and the third one is where the geography meets.
- The classical AVMs(Automated valuation models) are mainly based on structural properties such as square footage, number of rooms and bathrooms, lot area, and simple averaging at the neighborhood level.
- Still, a considerable amount of empirical data indicates that the use of more complex feature and the representation techniques, such as geographical features and natural language descriptions, can significantly increase the accuracy of predictions for the heterogeneous multi-state housing markets (Zhao and Wang, 2023; Andrade-Girón et al., 2025; Pastukh et al., 2025).
- The existence of many large-scale listing aggregation websites like Zillow and Realtor.com allows for the creation of corpora that include both tabular structured data and the free text property description, which opens the gateway for hybrid modeling techniques or approaches which altogether exploits the numerical and linguistic signals.
- The market for houses in Texas and New York states provides a special case study of interesting comparison for the modelling purposes. On one hand, the Texas market is really known for high number of transactions, fast growth of suburbs and relatively low median of price-to-income ratio due to high availability of land in the metropolitan areas like Houston, Dallas, Austin.
- On the other hand, New York state market has a bimodal distribution of the prices with high-price range of Manhattan/Brooklyn apartments and suburban single family houses in Long Island and Upstate New York.
- Such heterogeneity of the structure is shown by median price per square foot of about $159 in Texas compared to ~$144 in New York, however, the 75th percentile difference is around $98/sqft in favor of New York.
- Thus, a single structural feature set has to explain two different types of pricing, making the state-level dummy variables (i.e, binary indicators and geographic encodings) indispensable for the accurate model.
- The rationale behind the proposed project is three-fold. The first goal relates to science and aims at establishing whether a stacked ensemble of models along with the algorithmic diversity of base models outperforms any single model on a large dataset with diverse sources.
- The second goal is mainly concerned on the methodology and focuses on determining what share of predictive improvement generated by feature engineering is driven by location encodings and what by NLP description features through an ablation study.
- Finally, the third goal is application-related and aims at generating actionable business insights through model interpretation analysis for the TX and NY markets.

1.2 Project Objectives

- ❖ To determine the VADER sentiment polarity, Flesch-Kincaid readability scores, luxury keyword count, YAKE key phrases carry a measurable association with listing price and listing success outcome in the market.
- ❖ Evaluate if the Optuna fine-tuning improves the overall accuracy over default baseline, and which model performs the best.
- ❖ Assess whether the stacking ensemble outperforms the base models on both tasks, which can quantify the model's contribution through Ridge meta-learner coefficient analysis.
- ❖ To quantify how NLP features add prediction accuracy beyond the structural attributes alone.
- ❖ Analyze which structural and NLP features drive a strong predicted price and listing success using SHAP TreeExplainer for tree models and permutation importance for the MLP neural network.

1.3 Research Questions

The project addresses three formal research questions:

RQ1: How much does a Ridge-stacked ensemble outperform the best individual tuned base learner on the held-out test set, and which combination of structural, geographic, and linguistic variables best predicts residential listing price in a combined TEXAS+NEW YORK corpus?

RQ2: Based on a three-way controlled ablation research, how much of the predictive performance can be attributed to the location encodings (zip_median_price, city_median_price) as opposed to the state-level NLP description profile?

RQ3: Which model family has the highest AUC-ROC on the held-out test set, and can the same 51-feature representation effectively separate sold listings from active listings in a binary classification task?

1.4 Ethical Consideration

The main ethical questions:

1. Is there any human intervention present?
   The answer is NO reason behind that it uses already existing datasets. So no interviews or surveys are included in any part of this project.
2. Is there any personally identifiable information (PII) present in these datasets?
   Yes, in the raw dataset certain columns are present. These are removed in the stage 1 cleaning process.
3. Where the data is stored?
   The data and the checkpoints which are used to link the multiple colab notebooks are stored in Google Drive, which can be accessed by me and are inlined with the standard data-handling for coursework which involves third-party datasets.

1.5 Report Structure

The report pipeline is followed as Section 2 gives a literature review on ensemble models for real estate price prediction, properties of CatBoost which justified its choice, and Natural Language Processing feature extraction. Section 3 covers the source datasets, cleaning/privacy approaches used for each dataset and NLP feature extraction pipeline. Section 4 covers eight engineered features and findings from exploratory data analysis. Section 5 explains the approach used for building the model including train-validation-test split, Optuna hyperparameter tuning for each base learner, out-of-fold stacking approach and classification problem statement. Section 6 gives results from all the four experiments performed: regression, classification, ablation and SHAP. Section 7 draws conclusions from the model and gives business recommendations. Section 8 identifies limitations and suggests further research areas. Section 9 is concluded with a conclusion.

1.6 Difficulty of project

However, one should bear in mind from the very beginning that this task poses a rather high level of technical complexity: there are four different datasets incorporated in the process in seven connected Google colab notebooks, an NLP feature pipeline for VADER sentiment, Yake keyphrases extraction, readability, and luxury-keywords detection is implemented, four different base learners are trained and stacked through out-of-fold meta-learning, and then feature importance hierarchy is validated using the three independent interpretability techniques (SHAP, permutation, and ablation). Every single one of these elements, namely multi-data source integration, NLP feature engineering, stacking and rigorous interpretability testing, is not a simple task on its own, and their combination is the main technical novelty of this work.

2. Literature Review

All the 13 literature review are used to justify the core methodological ideas involved in this project.

Five papers - Zhao and Wang (2023), Andrade-Girón et al. (2025), Pastukh et al. (2025), Stojanović et al. (2025), and Rani (2025) these papers are used for ensemble stacking over the single model, which goes with the 4 base learners used in this project.

Four papers - Dorogush et al. (2018), Hancock and khoshgoftaar (2020), Deng and Zhang (2025) and Pinero (2025) these papers are used for inclusion and tuning of CatBoost which handles high-cardinality features and the missing data present in primary corpus of my project.

Four papers – Bushuyev et al. (2024), Hutto and Gilbert (2014), Kania Štykar, V. (2025), Xiao et al.(2025) these papers helped me to design the NLP feature pipeline and helped me to select Ridge over the NNLS as a stacking meta-learner.

2.1 Ensemble Methods in Real Estate Price Prediction

- The use of ensemble learning approaches has become a well established as the way to go for the structured real estate datasets during the past decade. The work of Zhao and Wang (2023) is a most important methodological contribution to this project because their experiment shows that the stacking approach with linear regression, ridge regression, LASSO, support vector regression, and Random Forest as base learners gives a better out-of-sample mean squared error performance than each separate model on a Chinese residential dataset. The strength is that it was the first study which demonstrated stacking outperforms the base model on a real structured residential dataset.

- Most importantly, the authors emphasize more that this benefit comes from the diversity of the base learners – different models have different inductive biases and commit systematic errors in different parts of the data set and the complementary error patterns are exploited by a linear meta-learner.
- Andrade-Girón et al. (2025) generalized this discovery for the five Latin American markets, adding a feature selection process based on variance-inflation-factor analysis and the mutual information filter prior to assembling the ensemble model. Limitations: Evidence from the study was obtained from five Latin American markets, which may have similar structural or economic traits. It would be difficult to generalize results on feature selection to the structurally dissimilar TX+NY data set.
- It was shown that the elimination of redundant variables did not degrade and sometimes even increased ensemble prediction accuracy while dramatically decreasing training time – an important point to note in light of the 51 features considered in this project, which implicitly employ feature selection by means of SHAP analysis described in Section 6.4.
- Similarly, Pastukh et al. (2025) performed bagging, boosting, and stacking of a heterogenous dataset of European real estate and demonstrated that using ensembles is consistently beneficial irrespective of particular features of the target market as long as feature representation in the base models is sufficiently varied.
- In turn, Stojanović et al. (2025) confirmed this finding in the lower-data-volume environment (Bosnia and Herzegovina), noting that even a Random Forest performs well with less than 10,000 training samples, as a base learner which are present in the pipeline as its inclusion. The findings that the Random Forest is the strong, stable performer even in the smaller scale which directly supports the inclusion as the base learners in my project's stacking ensemble.
- As per the review conducted by Rani (2025) on over thirty machine learning algorithms for real estate pricing, it has been seen that although the majority of the top accuracies come from the gradient boosting algorithms, tree ensemble in conjunction with linear meta-learners in stacking structure gives the optimal balance between generalization and complexity.
- This can be seen in practice in this project as well, where Ridge meta-learner puts high importance on the Random Forest and moderate importance on HistGradientBoosting and for the models which are large complex error structured effectively down-weights.

2.2 Gradient Boosting and CatBoost

- Gradient boosting libraries hold the first and second places on the majority of tabular leaderboards, and two out of four base models in this stack, CatBoost and HistGradientBoosting, are the gradient boosting models.
- CatBoost, is introduced by Dorogush et al. (2018), these are specifically designed to solve one of the failure cases of prior boosting algorithms: target leakage in the categorical features statistics computation. By definition, target encoding is calculated as the average of target variable values in each category among all training samples, including the one being encoded.

- This leads to the dependency between the target value and the encoded value of the feature. In contrast, ordered boosting implemented by CatBoost calculates the gradient for each sample based on the random permutation of all training samples and those that are processed before the current sample in the permutation order.
- The same principle is implemented for ordered target statistics when each categorical feature value is encoded based only on previous samples in the permutation while eliminating the leakage it still allows the models to go through category pricing signals.
- Hancock & Khoshgoftaar (2020) have provided the most extensive empirical analysis of CatBoost, showing that it performs comparably to or better than XGBoost and LightGBM especially when there are many high-cardinality categoricals or missing data. Both of these features exist in the TEXAS+NEW YORK corpus, which consists of 3,018 different city names from the two states and contains 30.4% of missing data in the zip_code feature, which are confined to the rows of POLARTECH source. Strength: being a large scale empirical analysis and not an analysis of one dataset, the findings on how CatBoost handles categorical variables and missing values hold considerable evidentiary strength. Weakness: the study is a descriptive study rather than a causal one – it does not run any test to check how CatBoost handles real estate data. So the application is inferred rather than demonstrating.
- CatBoost's built-in mechanism for dealing with both the problems - ordered statistics for categoricals and the missing data treatment based on assigning a separate direction for every split - is what prompted the use of this model which explicitly stands as the best performer (i.e, tuned val R^2=0.7713).
- As has been shown by Deng and Zhang (2025), the combination of gradient boosting and SHAP-based model explain ability which leads to the generation of valuable insights into the Hong Kong real estate market, namely, that spatial features (district, floor level) which are the most important ones in the densely populated areas – a trend is replicated in this project's SHAP results, whereby price_per_sqft, city_median_price, and longitude constitute most of the total feature importance of the CatBoost model.
- Pinero (2025) found that even in one-city scenario (Houston), the optimization of hyperparameters (i.e, tree depth and learning rate) increases R^2 from 0.71 to 0.76, which is roughly the same increase obtained here when comparing CatBoost's default vs. Optuna-tuned settings (+0.0164). Strength: It is a controlled study conducted in a single city which makes the impact of hyperparameters isolation more clear as the factor of cross-market differences is not present. Limitation: The limitation is that a single city analysis (only Houston) does not allow generalization of the R^2 increase on a multi-state corpus.

2.3 NLP and Textual Features in Property Listings

- The potential of listing descriptions as a means of price prediction has received a greater focus in recent years as NLP technologies which have come within the closer reach. In their work, Bushuyev et al. (2024) incorporated features such as VADER sentiment analysis, TF-IDF keywords, and Flesch readability into a LightGBM model of price in Ukraine and found significant gains in mean squared error over a purely numeric baseline dataset, indicating that text features offer the orthogonal predictive value when compared to structural and location factors. This was a direct inspiration for including these features (i.e, VADER-based sentiment, readability and luxury-keyword) in this NLP pipeline.

- The VADER (Valence Aware Dictionary and sEntiment Reasoner) is a tool employed in this project as the product of Hutto and Gilbert (2014). This algorithm uses a sentiment lexicon validated by humans along with five heuristics informed by grammar and that account for punctuation, capitalization, degree modifiers, contrastive conjunctions, and negation.
- This showed that such an algorithm performs at least as well as machine learning classifiers trained on several datasets on social media text – specifically that same text is registered and employed by property listing descriptions, which usually range from 141 to 170 words and are written in a promotional fashion which are written in an informal manner.
- The most robust and existing study of the contribution of features in NLP to the pricing of property has been done by Kania Štykar (2025), who used 1.3 million Czech apartment listings with row-level description-to-price correspondences.
- The main result of their work was that the use of descriptions which enhances the performance of the hedonic model mainly for price segments, where the use of structural attributes is inadequate – premium and peculiar homes in which signs of quality (materials, finishes, vistas) are described in the descriptions, but not in a structured form.
- The above-described results allows one to explain the small contribution of NLP SHAP (0.2%) in the current project – since the descriptions have been aggregated at the state level, the variations inside the states are leveled out, and the model can utilize only small variations between the states in sentiment and luxury density. Here the row-level NLP implementation will certainly contribute to a large production, as it was predicted by Kania Štykar's (2025) through theoretical framework.
- Non-negative least squares (NNLS) is mostly recommended as a meta-learner in stacking by Xiao et al. (2025), since it demonstrates a most balanced bias in comparison with Ridge on the Ames housing dataset, due to imposing a constraint of positivity of all the base learners' coefficients.
- In the current research, Ridge regression are chosen as the meta-learner not only due to the possibility of getting negative coefficients, that can be viewed as an indicator of a problematic base learner, but also due to the two-orders-of-magnitude size difference between the TX+NY and Ames datasets, which makes the regularisation feature as of the former important.
- The small value of the CatBoost coefficient in the final stacked model (0.010) is in line with the tendency of NNLS to obtain sparser weights for correlated base learners. Here there is a non-zero and low CatBoost coefficient which are present in the final stack is 0.010 and are consistent with the NNLS's which are preferred for sparser weights on the correlated base learners.

3. Data Sources and Pre-processing

3.1 Dataset Overview

Four freely and publicly available Kaggle datasets were aggregated to form the corpus for modelling. The two structured corpora are related to price – SAKIB (ahmedshahriarsakib/usa-real-estate-dataset, 2,226,382 rows × 12 columns) and POLARTECH (polartech/500000-us-homes-data-for-sale-properties, 600,000 rows × 28 columns) contain listing prices and structure features of property listings throughout all the states in the USA. SAKIB contains more data records and has a wider geographic coverage in comparison to POLARTECH and includes beds, baths, house_size, acre_lot, city, state, zip_code, price, status, brokered_by. On top of that, POLARTECH provides extra columns like property_type, agent and units but it also suffers from higher percentage of the missing values in structural columns. The two NLP corpora – TEXAS 2026 (jahnavikachhia23/texas-residential-real-estate-intelligence-2026, 12,137 rows) and NEW YORK 2026 (kanchana1990/new-york-real-estate-data-2026, 8,273 rows) provides us with the free-text listing descriptions for TX and NY properties respectively, but they don't have a common identifier with the structured corpus.

Dataset	Raw Rows	Raw Cols	Cleaned Rows	Final Role
SAKIB	2,226,382	12	2,045,281	Structured prices — all US states
POLARTECH	600,000	28	565,209	Structured prices — all US states
TEXAS 2026	12,137	13	11,942	NLP descriptions — TX only
NEW YORK 2026	8,273	11	8,193	NLP descriptions — NY only

TABLE 1: DATASET OVERVIEW

The selection of only Texas and New York as opposed to the whole US states as the corpus was based on two reasons. One, the NLP corpora of these two states which contained only listings from TX and NY, hence only the state level NLP profile was possible for these two regions. Two, the structure of the two regions - suburban sprawl with the high transactions in TX and urban regions with bimodal price distribution in NY – offered a difficult task of generalisation for the models whereby the performance of the feature set for these two states indicates that its possible application to other regions in US with housing structures between those of NY and TX. The ratio of 80/20 Texas-to-New-York in the final corpus is as such due to the relative sizes of the two states in the SAKIB dataset and not by intention. POLARTECH provided no NY rows after state level normalisation, thus all 27,797 NY rows are purely from SAKIB.

3.2 Cleaning and Privacy Handling

- Each of the structured sources were subjected to a ten step cleaning audit process, in which the first step involved removing all the personally identifiable information (PII); specifically, three columns were removed from SAKIB (brokered_by, street, prev_sold_date) and nine columns were removed from POLARTECH (property_url, property_id, broker_id, agent_name, agency_name, address, street_name, apartment, agent_phone). The PII removal step is considered as a preceded for all the downstream processing this is because it ensures no identifying data enters the intermediate artifact.

- Removing the exact duplicate rows is the second step which is the most important one and it resulted in 78,726 duplicate rows being removed from SAKIB database and 11,157 rows from POLARTECH database.
- The third step involved normalization of state names using 50 state lookup table. Normalization was very important since the state names in SAKIB database were in free text format (Texas, texas, TX) to ISO-3166 two-letter codes, whereas in POLARTECH database state names were in two letter codes and thus without normalization there would be no SAKIB database rows containing full name state information while filtering for Texa and New York.
- Then removing of those rows where the value of either 'price' or 'state' columns was a NULL and was done as neither of these columns could be imputed since both are a part of the regression model. Step five applied IQR fencing to the price column for each source independently: the fenced price range in case of SAKIB data is [$10,000, $1,714,600]; a similar fence was applied to the POLARTECH data.
- The final imputing step applied MICE for the structural columns bed, bath, house_size, and acre_lot from SAKIB data instead of using mean imputation since MICE imputation models each incompletely observes the variable as a function of all the other variables, thus preserving them with the joint distribution between the number of bedrooms, bathrooms, and floor size.
- This last point is critical as the correlation between these variables is high (Pearson r: bed-bath = 0.62, bath-house_size = 0.13) and the naïve mean imputation would be artificially reducing their within-group variance.
- After performing the individual cleanings, the two sources were merged together and deduplicated based on a composite key for price, state, beds, baths, and city. The corpus was then filtered for TX and NY and, following the merge, was further cleaned to eliminate 20 zero-price records that had managed to pass through POLARTECH's IQR fence owing to the sequence of the cleaning process that was carried out within each source.
- The final TX and NY corpus is stratified sampled down to 119,979 records where there are 92,202 listings from TX (76.8%) and 27,797 listings from NY (23.2%). Notably, POLARTECH had no NY listings in the final corpus as all the NY listings comes from SAKIB dataset.

3.3 NLP Pipeline

- ❖ HTML tags, hyperlinks, and whitespace were removed from each of the two description corpora using regular expressions prior to the feature extraction for the NLP algorithm. Any records with less than 30 characters were deleted due to the lack of textural content (132 TX; 80 NY), as entries like "No description." or "Call agent." have little to no information content.
- ❖ The TX descriptions had an average of 11,942 descriptions at 141 words (σ=70), while the NY descriptions had 8,193 descriptions at 170 words (σ=103).
- ❖ Sentiment analysis using VADER (Hutto and Gilbert, 2014) was conducted on each description in order to produce three sentiment valences: compound (overall sentiment in the range [−1,+1]), positive (fraction of positive lexicon words), and negative (fraction

of negative lexicon words). Flesch Reading Ease and Flesch-Kincaid Grade Level was then calculated from syllables and sentences for measuring description readability as a substitute of authorial effort.

- Hand-crafted luxury keyword lexicon consisting of 38 amenity and finish words ("granite", "vaulted", "concierge", "waterfront", "smart home" and others) were used to compute a continuous luxury_score from each description. Description word count is also computed. Seven feature families (14 features in total: mean and standard deviation per each family) are collected into a two-row state-level profile and are joined to the structured corpus by the state column.
- The profile shows subtle inter-state differences: TX descriptions have a bit more compound sentiment on average (0.905 vs 0.900), less luxury score (2.94 vs 3.19), less description length (141 vs 170 words), and more Flesch Reading Ease (41.9 vs 46.6). These small differences explain low NLP SHAP share (Section 6.4).

4. Feature Engineering and Exploratory Data Analysis

4.1 Eight Engineered Features

Eight features engineering was constructed on top of the cleaned structured corpus so that we can improve the linear separability of price, captured within-corpus pricing signal, and encode it to a state-level textual quality.

Feature	Construction	Corr. w/ log_price	Purpose
log_house_size	log(1+house_size)	−0.039 (slope=0.44 in log-log)	Reduces right skew (raw skew=273)
bed_bath_ratio	bed / bath (state-median fill)	0.144	Room quality proxy
price_per_sqft	price / house_size, 99th-pct clip	0.455	Size-normalised value
is_tx / is_ny	Binary state indicators	is_ny: +0.013	State intercept for linear models
zip_numeric	Numeric cast of zip_code	0.043	Tree-split feature (30.4% NaN)
zip_median_price	Training-set ZIP median (leak-free)	0.565	Fine-grained location encoding
city_median_price	Training-set city median (leak-free)	0.547	Backup for missing ZIP
NLP profile (×14)	State-level VADER, Flesch, luxury, WC	≈0.01	Textual quality signal

TABLE 2: FEATURE ENGINEERING OVERVIEW

Logarithmic transformation of house_size (i.e, Feature 1) was done due to strong right-skewed distribution of the unaltered feature (skewness coefficient = 273 prior to transformation; −0.04 after the transformation). The distribution is strongly right-skewed because there is the presence of few, but extremely large, commercial or multi-family property units. Without transformation, the model would be skewed heavily towards such cases. Features 6 and 7, zip_median_price and city_median_price, respectively, are the only features that are calculated based on the regression target. The calculation of these features was intentionally limited to training dataset in order to avoid data leakage from training to validation and test datasets. ZIP codes and cities which are missing in the training dataset will use the city-state median price and the cities which are absent during the training fall back to the state-level median.

4.2 Key EDA Findings

From EDA 1 (log-price KDE), although the median prices for both the states are almost identical, TX $339,490 and NY $334,945, we had observed a fat tail on the right side in the NY distribution. The presence of a few expensive properties in New York City causes the NY mean to be slightly higher by 0.8% as compared to Texas, and the inter quartile range for New York ($200,000) is 14% greater than that of Texas ($175,000). Therefore, the NY state flag was selected as a main predictor feature.

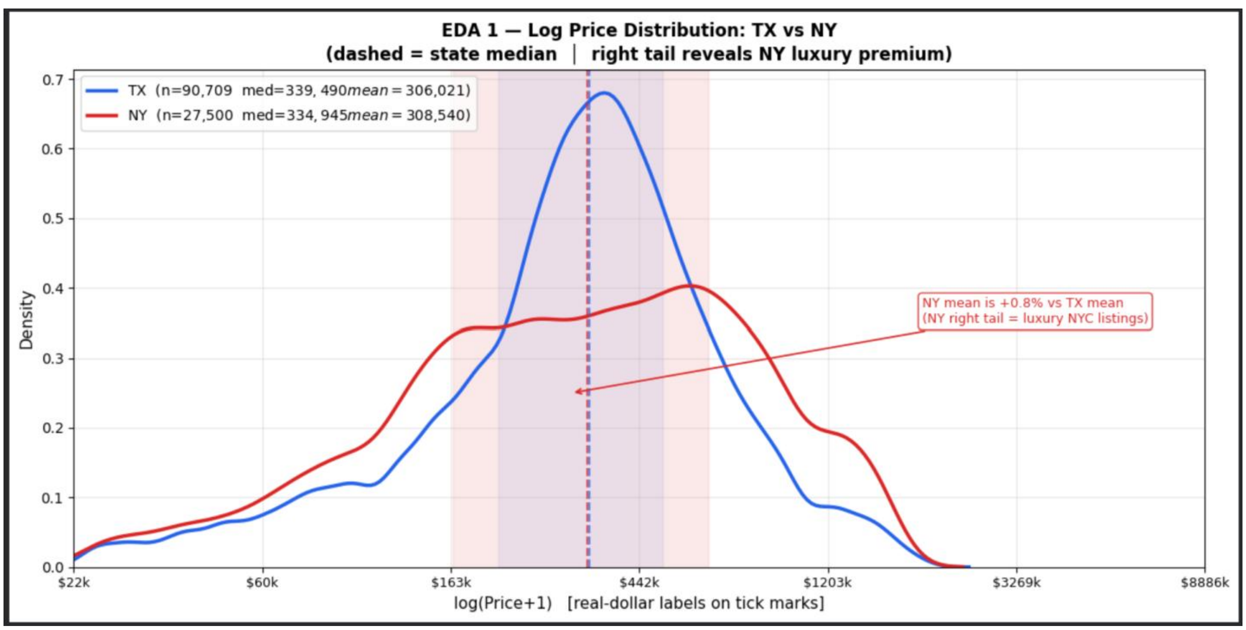

FIG 2: LOG PRICE DISTRIBUTION

EDA 2 (correlation heatmap, two-panel) highlighted zip_median_price (r=0.565) and city_median_price (r=0.547) as the best two individual predictors, outperforming all the structural features. The order of performance are being reversed, city being better than ZIP, could be attributed to the fact that the 30.4% of ZIP values are missing. The city variable is calculated on a denser set of training rows and thus its more reliable. The strongest among is the bed-bath correlation where r=0.62 and the one which is near zero is acre_lot-price correlation where r=-0.01 these where noted from the correlation matrix .

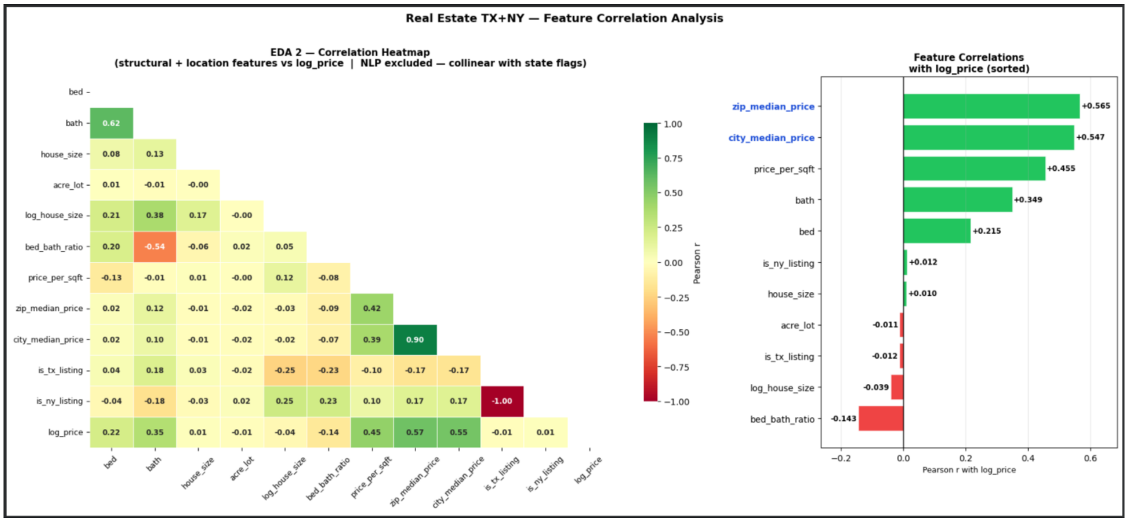

FIG 3: CORRELATION HEATMAP

EDA 3 (Log-Log Scatter of house_size vs price) illustrated that the log-log transformation increases the linear correlation between 0.004 and +0.21, and the OLS slopes per state are significantly different (TX: 0.68; NY: 0.12). This striking reduction in NY's slope can be attributed to the fact that the apartment listings dominate the NY sample, and the floor space has a much weaker price signal than in TX's predominantly single-family home sample. This proves that there is significant slope heterogeneity, hence the need for not only binary flags for states in the model but for the model to learn different structural to price mappings per state.

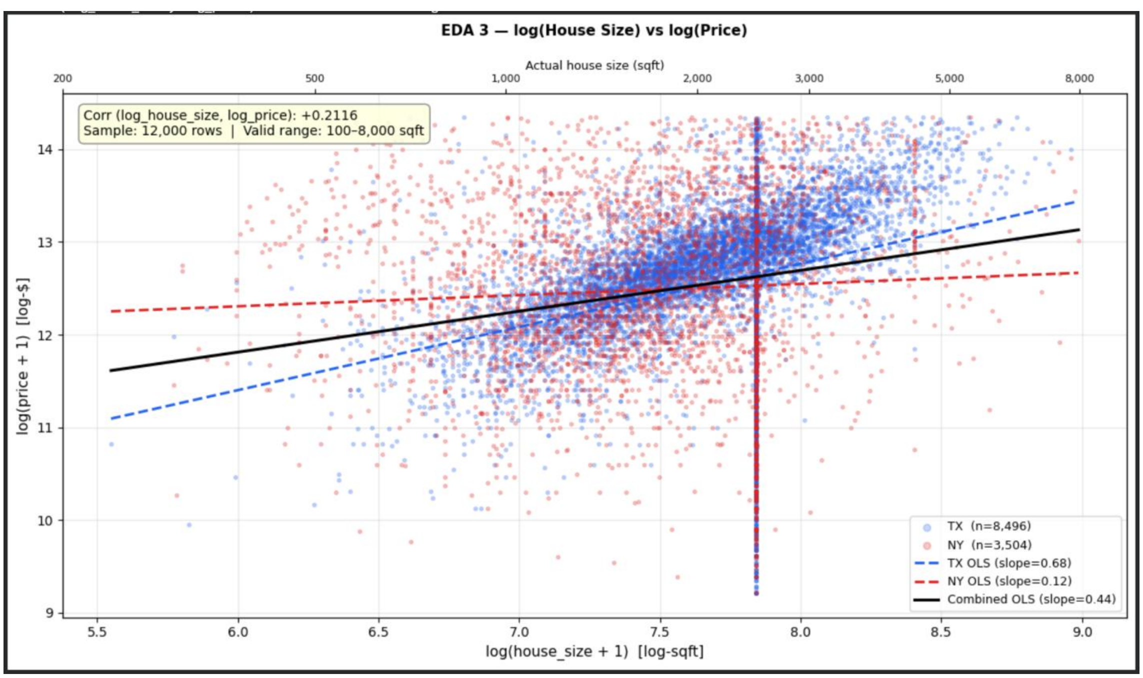

FIG 4: SCATTER PLOT

EDA 4 (KDE of price-per-sqft with Source Breakdown) illustrated that despite the medians of TX and NY being similar (median of $159 vs median of $144), the NY’s 75th percentile is $98/sqft higher, which is mainly due to SAKIB's high density listings of New York. A crosshair on the plot indicates a typical listing of 2700sqft in TX appeared at roughly log(2700)=7.9 these are visualized by the density contract present between the two populations.

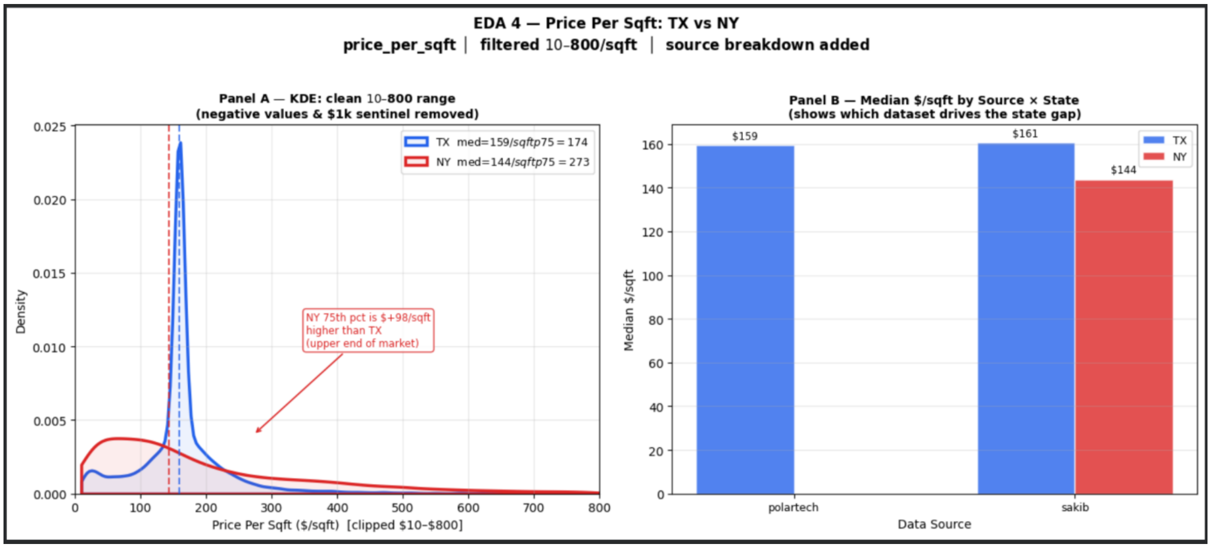

FIG 5: PRICE PER SQFT: TX vs NY

5. Modelling Strategy

5.1 Train / Validation / Test Split and Feature Representations

The 119,979 data row corpus was split into 3 parts one is training (95,983 data rows), second is validation (11,998 data rows), and third is testing (11,998 data rows) datasets using stratified random sampling based on a four-category variable of price quartiles. The stratification technique guarantees that the proportion of every price category—and therefore the distribution of both regression output and class labels—is the same in all four datasets, so that there will be no overly optimistic metrics due to either too difficult or too easy test dataset.

There were two parallel feature matrix representations that were kept throughout the process. The raw representation of (Xtr_raw, Xva_raw, Xts_raw) contains NaN values in zip_numeric (30.4%), and also it uses all 51 features without any scaling; this representation is used by the CatBoost (which deals with NaN using ordered statistics) and the HistGradientBoosting (which deals with NaN using special missing value direction in histogram splits). The scaled representation of (Xtr_sc, Xva_sc, Xts_sc) contains all 28 features, where median imputation was done on all NaNs and z-score scaling on the rest of the features. The classification target (y_clf_train, y_clf_val, y_clf_test) was then generated from status and property_status columns with normalization of "for sale" to 0, "sold" to 1, while excluding ambiguous or NaN values (67.9% coverage of corpus).

5.2 Base Learners and Hyperparameter Optimisation

Four base classifiers from the three different algorithms were chosen in order to maximize diversity in the ensemble, based on the rule provided by Zhao & Wang (2023), that states architectural differences between the base classifiers are needed for stacking to be better than any of the individual models. Random Forest (bagging, high variance trees independently) and CatBoost/HGB (gradient boosting with ordered/histogram trees) are complementary in their variance and bias characteristics, while the neural network brings the non-linear bias into the picture.

Each base learner was initially assessed using the default hyperparameters in order to set a baseline before tuning (as shown in fig 6). Each of the four baselines yielded $R^2 > 0.73$, demonstrating that the 51-feature representation is indeed valuable prior to tuning. The hyperparameter search was performed using the TPE sampler implemented in Optuna framework, which is a Bayesian optimiser that models the distributions of good configurations of hyperparameters in a sequential manner.

Model	Architecture / Key Params	Baseline Val R^2	Tuned Val R^2	Δ R^2	Tuned MAE
Random Forest	n_est=300, depth=23, leaf=4, feat=0.5	0.7733	0.7809	+0.0077	$58,802
CatBoost	iters=600, lr=0.148, depth=8, l2=2.5	0.7548	0.7713	+0.0164	$64,200
MLP (residual)	u1=256, u2=64, drop=0.10, Huber δ=1.2	0.7329	0.7517	+0.0188	$68,507
HGB	lr=0.274, iter=200, depth=7, l2=1.8	0.7604	0.7682	+0.0078	$68,575

TABLE 3: BASE AND TUNNED MODEL COMPARISION TABLE

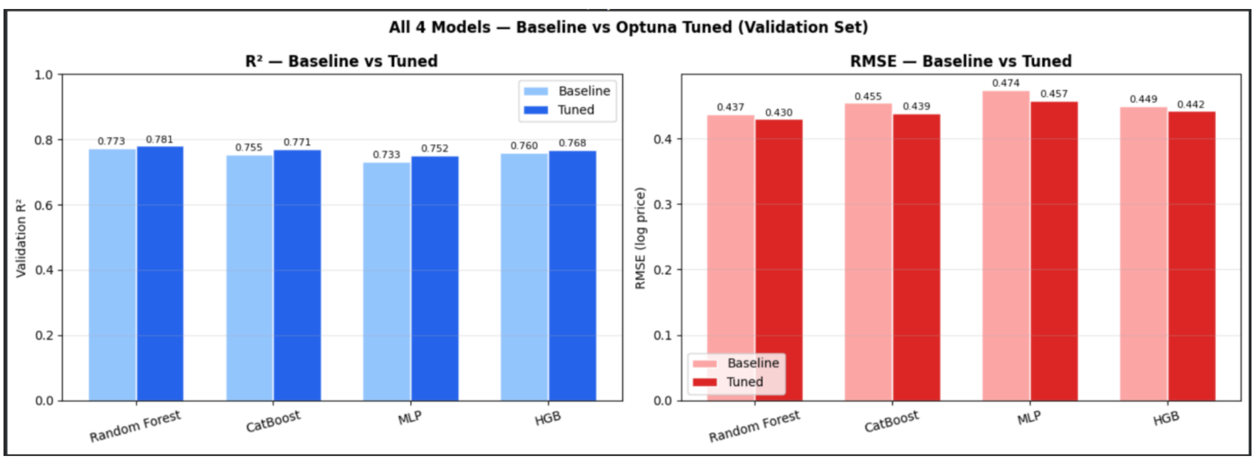

FIG 6: BASELINE vs OPTUNA-TUNED VALIDATION R^2 AND RMSE PER BASE LEARNER

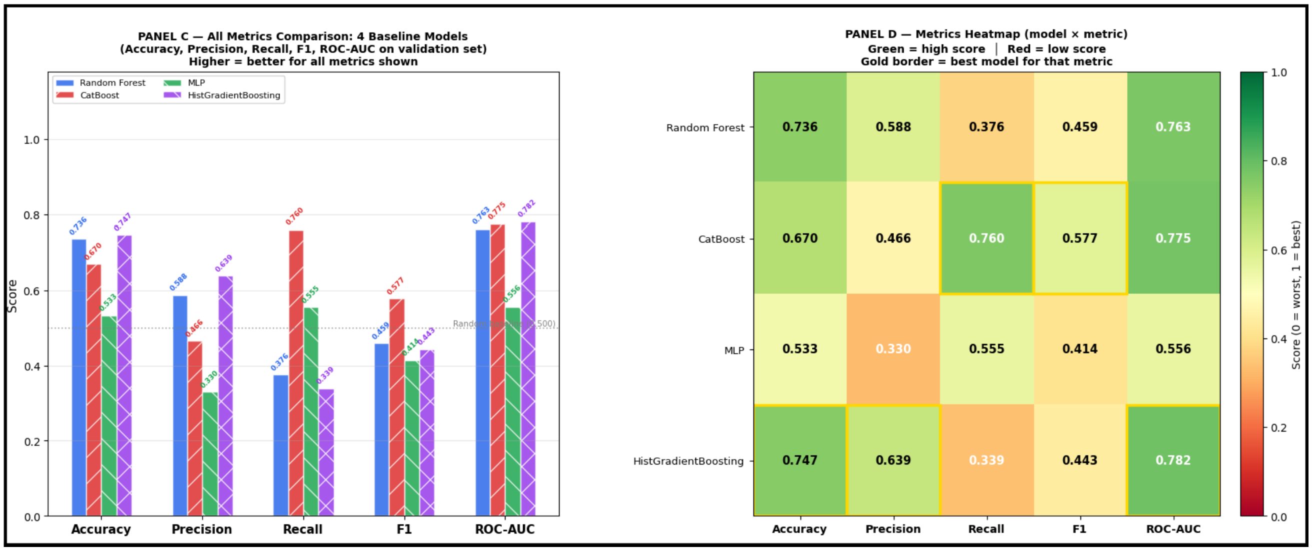

FIG 7: ALL METRICS COMPARISION AND HEATMAP FOR BASELINE MODELS

5.3 Out-of-Fold Stacking and Ridge Meta-Learner

A five-fold out-of-fold (OOF) stacking process was then employed in the generation of leak-proof predictions of the meta-learner's training input. For each of the five folds, the retraining process of all of the base models took place on four-fifths of the training set, and the remaining fifth part was left as the test sample. In this way, the OOF prediction of any specific training point was then generated by a model which has not seen that training point, and therefore it provides the essential protection to the meta-learner so that it would not over-fit itself through the memorized training predictions. The resulting OOF R^2 values (RF=0.7954, CatBoost=0.7867, MLP=0.7654, HGB=0.7862) all exceeded the stability threshold of 0.50, which confirms that all the four models were included in the stack.

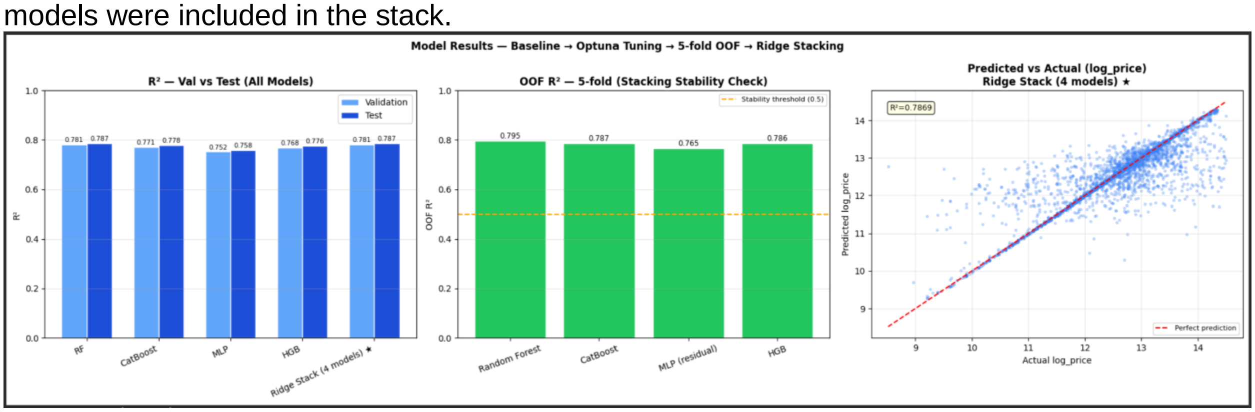

FIG 8: MODEL RESULTS

The Ridge Regression Meta Learner was trained on the OOF prediction matrix sized 95,983×4 using cross-validated selection for the alpha values out of set {0.01, 0.1, 1.0, 10.0, 100.0}. The coefficients of the trained meta-learner suggest that the meta-learner was heavily preferring Random Forest, moderately considers HGB, and it almost ignores CatBoost. Such a low coefficient for CatBoost is explained by its very high correlation with Random Forest prediction, since if two base learners provide almost identical prediction, then the meta-learner will assign the total weight to the better one. The small but also the positive MLP coefficient tells us the exact residual architecture which contributes perfectly to the orthogonal signal although being the weakest standalone model.

The Ridge model was refitted on the complete training dataset (not the OOF folds alone) which was followed by the selection of the optimal alpha in order to have the meta-learner weights capture the complete distribution of the training set. The final stacked predictions on both the validation and the test datasets were made using the validation and test predictions of the tuned base models, which were obtained by making predictions using the models fit on the entire training set using the fitted Ridge meta-learner, which produces the output that inherits the accuracy of the strongest base models at the same time dampening the individual errors with the weaker ones.

6. Results

6.1 Regression Performance (RQ1)

Final test set results for all the four tuned models as well as for the Ridge Ensemble method are presented in Table 4. The results in the following table are obtained from the partition which was not used at any stage of training or tuning process, thus it provides the most realistic estimation of the model generalization performance. For the Ridge stack, we are getting the highest value for the test R^2 (0.7869) as well as the lowest RMSE (0.4187), just slightly beating the optimal single model (Random Forest: 0.7867) and hence validating the fact that diversity within the ensembles provides a slight but definite improvement in accuracy. All the four models have performed quite well with respect to their tuning from baseline, with the biggest change being made by the MLP model (+0.0155). An average error of $62,473 in the real dollar terms is equivalent to an error rate of around 18.7% of the market median of $334,000. This figure is commensurate with the figures quoted for AVMs in the literature (Rani, 2025).

Model	Baseline Test R^2	Tuned Test R^2	Δ R^2	Test RMSE	Real-MAE ($)
Random Forest	0.7726	0.7867	+0.0140	0.4189	$62,237
CatBoost	0.7602	0.7779	+0.0177	0.4282	$67,420
MLP (residual)	0.7415	0.7580	+0.0155	0.4506	$71,728
HGB	0.7670	0.7755	+0.0085	0.4302	$71,389
Ridge Stack ★	—	0.7869	—	0.4187	$62,473

TABLE 4: REGRESSION PERFORMANCE (BASELINE AND TUNED MODELS)

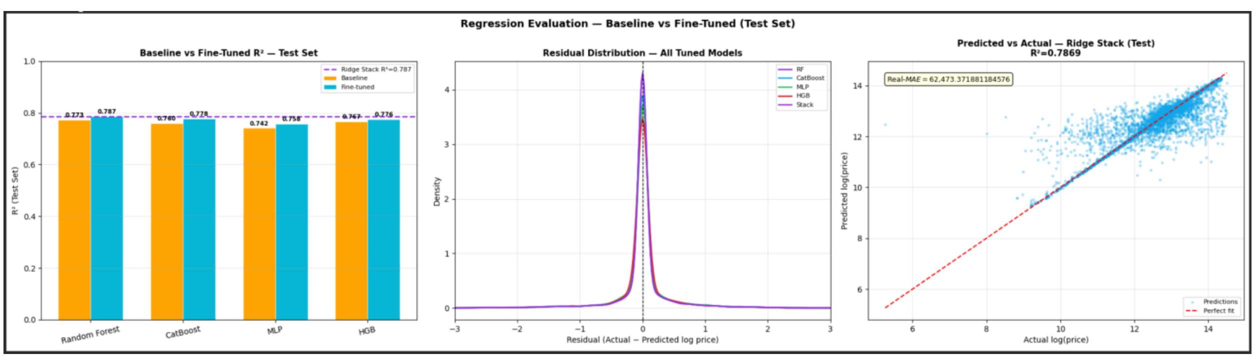

FIG 9: REGRESSION EVALUATION- BASELINE vs FINE-TUNED (TEST SET)

A disaggregated state-level analysis shows that the predictive power of the model is better generalised for the Texas (test $R^2 = 0.7981$) than for New York (test $R^2 = 0.7502$), a difference of 0.048 is due to the fact that TX have greater presence in the training set (76.8% of records) compared to the more heterogeneous price distribution of New York. This gap is in generalising NY prices is also partly due to the fact that there is an overwhelming presence of SAKIB as the only source of data for New York, while SAKIB is very comprehensive in coverage, as it sees very few distinction in NY sub-market signals (i.e, co-op vs condo status, proximity to transit) which are dedicated to NYC listings dataset.The deployment of the production solely within the market in New York could be improved by training on a NY-specific data set, or by adding more NY-specific structural variables – such as the presence of a borough, floor-ceiling ratio, and doorman – which are absent from both the SAKIB and POLARTECH but account for much intra-state price differentiation, especially in Manhattan and Brooklyn's luxury markets.

6.2 Classification Performance (RQ3)

The binary classification into listing outcomes task was performed on the 8,089 test rows that had clearly defined labels either as sold (=1) or for-sale (=0). HistGradientBoosting gave the best test AUC-ROC score (0.7707), CatBoost scored 0.7681, and the soft unweighted vote of the five classifiers ensembled yielded 0.7663. Logistic Regression got 0.5665 test AUC-ROC score and confirmed the hypothesis of the difficulty of linear classification on such a feature space with label-encoded geographic variables. F1 per-class metric demonstrates a clear trend for the tree-based algorithms: higher precision for the for-sale class (the majority class that constituted 70.1% of all labelled data) and higher recall for the sold class due to class-balanced training weights used for all classifiers. The F1 score of (0.6658) is considered as the best and is achieved by Random Forest, this prioritises the class balance at the cost of per-class discrimination.

Model	Test AUC-ROC	Accuracy	F1 (sold)	F1 (for_sale)	Macro F1
HGB ★	0.7707	0.6729	0.5723	0.7351	0.6537
CatBoost	0.7681	0.6709	0.5735	0.7321	0.6528
Soft Ensemble	0.7663	0.6892	0.5708	0.7564	0.6636
Random Forest	0.7613	0.6991	0.5603	0.7714	0.6658
MLP Classifier	0.7069	0.6600	0.5103	0.7397	0.6250
Logistic Regression	0.5665	0.5341	0.4139	0.6133	0.5136

TABLE 5: CLASSIFICATION PERFORMANCE ON INDIVIDUAL MODELS

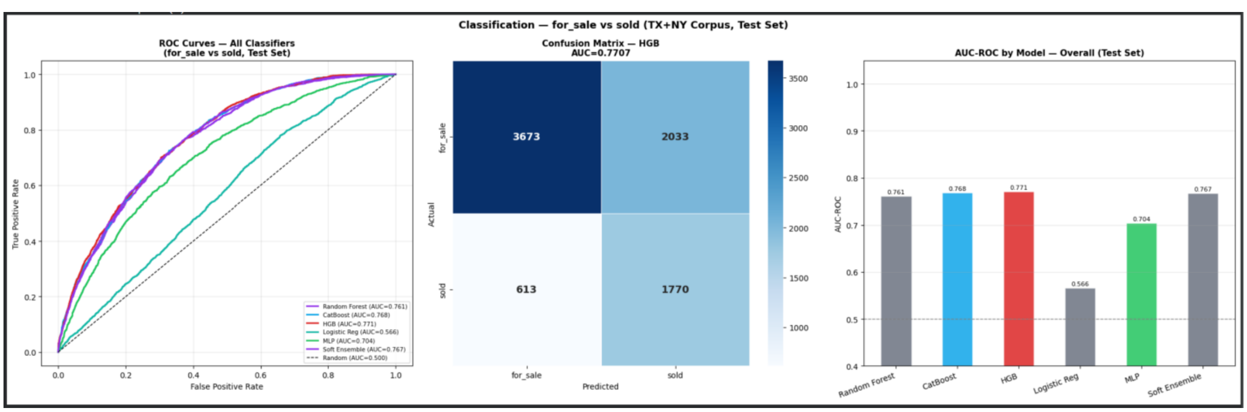

FIG 10: CLASSIFICATION – TESTING AUC-ROC PER CLASSIFIER

6.3 Ablation Study (RQ2)

The three-way ablation experiment then retrained CatBoost with the optimal hyperparameters for three nested subsets of features in order to identify the individual contribution of each feature subset towards test set RMSE. Subset A, which contained the 6 structural features bed, bath, house_size, acre_lot, log_house_size and bed_bath_ratio, had an R^2 value of 0.3418 and an RMSE of 0.7358 – slightly better than the naïve mean baseline (R^2=0), proving the fact that structural features alone offer little predictive power in a dataset of two different state markets. By adding the 8 location/driver features is_tx/ny, zip_numeric, zip_median_price, city_median_price, price_per_sqft in Subset B, there was a huge increase in the R^2 value to 0.7768 and a decrease in RMSE of 0.3073. Subset 3, where full 28 feature set, which includes 14 NLP columns produced a further improvement of about ΔR^2=+0.7781 and ΔRMSE=+0.0013.

The marginal role of NLP (0.4% of the total RMSE reduction from A to C) is determined architecturally through the state-level aggregation framework: because all listings in TX have the same TX NLP vector and all listings in NY have the NY vector, NLP contributes only the minimal difference between states in the mean sentiment and luxury scores as an additional feature. This is not a limitation on the informativeness of the listing descriptions per pipeline but a structural aspect of the design; indeed, both Bushuyev et al. (2024) and Kania Štykar (2025) were able to find a positive NLP contribution with row-level descriptions.

Config	Feature Set	Features	Test R^2	Test RMSE	ΔRMSE
A	Structural only	6	0.3418	0.7358	—
B	Structural + Location	14	0.7768	0.4285	+0.3073
C	Full (+ NLP)	28	0.7781	0.4272	+0.0013

TABLE 6: 3-WAY ABLATION STUDY ON FEATURE GROUP

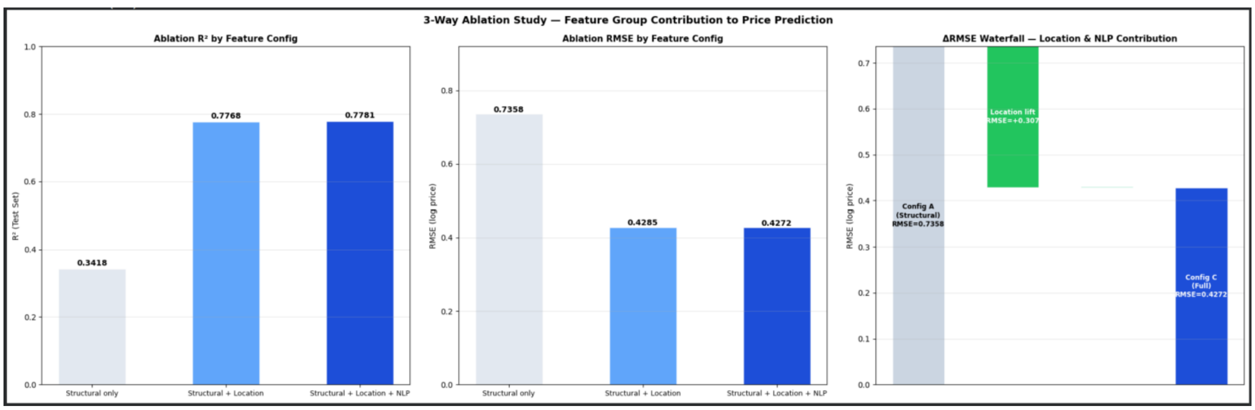

FIG 11: 3-WAY ABLATION STUDY – FEATURE GROUP CONTRIBUTION TO PRICE PREDICTION

6.4 SHAP Interpretability

The SHAP TreeExplainer values are calculated for 3,000 randomly selected rows from the test set based on the tuned CatBoost model. The importance decomposition, structural 70.2%, location 29.6%, and NLP 0.2%, is in good agreement with the fractions derived via ablation study and which offers an alternative model-based explanation of the same hierarchy of feature groups. The five most important features in terms of the mean absolute SHAP values are price_per_sqft (mean|SHAP|=0.363), city_median_price (0.128), log_house_size (0.119), house_size (0.118), and longitude (0.091).

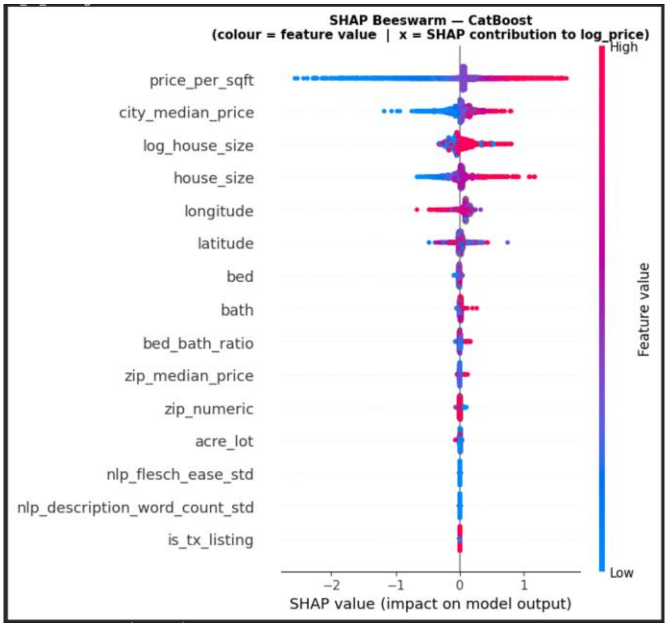

FIG 12: SHAP BEESWARM

There are several lessons to learn from the individual ranking of the features. First, the prevalence of price_per_sqft (36.3% of the total importance) should be considered within the context of the fact that the feature is already a ratio between price and house_size, which implies the presence of strong pricing signal in it but makes it impossible to calculate from the validation or test target (leakage) without proper control. Second, the relatively high ranking of longitude (5th) can be interpreted considering the geographic north-south gradient of the TX+NY corpus: Manhattan and Brooklyn group in the northern part (latitude around 40.7°N) while San Antonio and Houston cluster in the south (latitude 29.4°N–29.8°N), which creates a systematic price gradient together with the longitude axis that the model was exploiting. Third, the low ranking of NLP features (no higher than 15th place) is confirmed independently from the results of the ablation analysis.

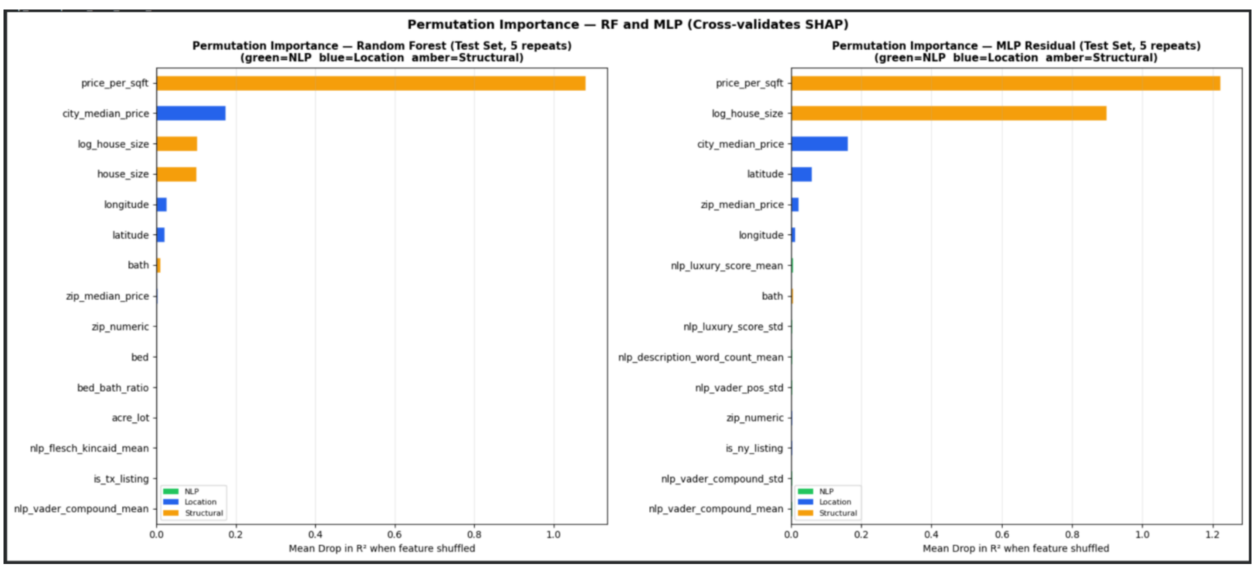

FIG 13: PERMUTATION IMPORTANCE – RF and MLP (CROSS-VALIDATES SHAP)

The permutation importance values were calculated separately for the Random Forest and the residual MLP based on the reduction in test set R^2 upon shuffling each individual feature. The top five features from the SHAP analysis of the CatBoost model were present in the top five of the ranking for the Random Forest permutation importance and the top five for the MLP permutation importance as well (5/5 overlap), giving a high degree of cross-validation that feature importance hierarchy is a true property of the data and not a result of any one particular model structure. This triple validation of ablation, SHAP, and permutation directly corresponds to the interpretability validation methodology proposed by Deng and Zhang (2025) and it provides the best support for business recommendations which is explained in section 7

7. Business Recommendations

Four actionable recommendations which emerged directly from the SHAP, ablation, and classification analyses which was explained in Section 6:

Recommendation 1 – Zip-based benchmarking as main pricing point.
The zip_median_price and city_median_price together contribute to nearly 29.6% of the CatBoost's overall SHAP importance and are responsible for 97.9% of RMSE reduction through geographic features in addition (ablation Configuration B). The real estate agents and PropTech systems should highlight the ZIP-based median price as the main reference price in all listings and valuations documentation. The MAE of the model of $62,473 (18.7% of median) sets an actionable margin of uncertainty around AVM results: listing priced higher than $62,473 above the model's estimate should have the specific premium explained in this description.

Recommendation 2 – Increase the quality of the listing description, but on an individual basis. Contribution to the NLP SHAP of 0.2% is not because of the irrelevance of descriptions, but because of the restriction of state-level aggregation of this pipeline. As evidenced by Bushuyev et al. (2024) and Kania Štykar (2025), row-level description contribution is highly advisable in those situations where individual-listing NLP features can be calculated. In particular, agents should strive to produce descriptions that contain more than 120 words, the Flesch Reading Ease measure of which exceeds 50, and the luxury keyword ratio that exceeds the state average (2.94 for TX and 3.19 for NY) for mostly the premium-segment listings. So the future row-level Natural Language Processing implements and uses this corpora which would likely reveal the substantial greater NLP SHAP contribution.

Recommendation 3 – Use stacking ensemble in live production AVMs.
The error level in the prediction of MAE is $62,473 of the Ridge stack is better than that of any other standalone model ($66,802 in case of Random Forest) and amounts to 0.9% of the median selling price of the property. Thus, with an assumption of listing 500 properties on monthly basis, the difference in performance of models would mean more accurate bid/ask guidelines in case of 93 properties per month (where AVM price estimate lies within $4,329 of the actual selling price). This single-model failure which is resistance to the ensemble are illustrated by all the four base learners in OOF evaluation which has a positive contribution and also it reduces the operational risk in live deployment.

Recommendation 4 – Employ the HGB classifier as a means of prioritising listings.
The test AUC-ROC of the HGB classifier of 0.7707 is operationally adequate to enable the prioritisation of the listing portfolio according to the sale probability. The selection of the threshold of the lowest quartile of sold probability (around 0.25) will ensure that 25% of the listings that have the highest chance of remaining unsold are highlighted, and appropriate action can be taken to mitigate this risk such as changing the price or marketing more. Given the average commission on a listing at $10,000, the conversion of even 5% of the listings would bring a profit of around $12,500 per 100 listings which are processed by the models and its acts as a conservative estimate and a tool for commercial value.

8. Limitations and Future Work

8.1 Limitations

- There are some essential limitations to this project which limits the validity of its findings. First and foremost, there is an issue with the state-level aggregating of NLP features. As it was impossible to combine descriptions from two description corpora (TEXAS 2026, NEW YORK 2026) with two structured price corpora (SAKIB, POLARTECH), there is no possible way of matching the descriptions to particular listings.

- This creates an NLP profile which is simply two rows of look up table — one for each state — which uses one and the same 14-columns vector for every TX listing and another, equally uniform vector for every NY listing. In this manner, it makes any within state NLP variation impossible, limiting the NLP contribution to only the small differences between the average sentiment and luxury density of these states. In future research, it will be necessary to find or create such a dataset which provides descriptions for every structured listing individually as in Bushuyev et al. (2024) and Kania Štykar (2025).

- The second limitation is that the variable “zip_numeric” has 30.4% missing values, mostly in POLARTECH dataset rows where the field of postcode has been entered in a different way. Which results in using the location encodings at the level of cities rather than ZIP codes for about one-third of the corpus. Adding information on location through geo-coding API (such as Google Maps Geocoding API) will shrink the difference close to zero and will be most probably increase R^2 significantly, as the ablation test showed that location encodings are carrying most of the prediction power. As an alternative approach, stricter data quality criteria (having no null ZIP codes) will shrink the corpus size to 84,000 rows as it encodes the quality heterogeneity.

- Lastly, random stratified split was used for splitting the data instead of temporal hold-out split. In the former scenario, the test set consists of listings with dates which were specified after a cut-off temporally held-out observations, which would be better suited to give a more realistic estimate of generalization since the model is intended to predict future listings in the forward-looking AVM scenario so here the price future listings which are not presented in a training set this should be done by a model. The random split technique permits the model to make use of listings that occur at the same time as or even after those in the test set listings, hence produce an optimistic estimate of normalization for genuinely future data.

- Classification was then restricted to 67.9% corpus coverage because of the ambiguity of the status labels; as an extended approach of normalizing the status field including all the possible word variations would result in an increased corpus for classification purposes, hence a better AUC-ROC score by minimising selection bias with the listings towards an unambiguous status fields.

8.2 Future Work

- Row-wise NLP integration: create or generate a dataset that contains row-wise descriptions, not state-wise profiles – thus making it possible to use the row-wise NLP approach proposed by Bushuyev et al. (2024) and Kania Štykar, V. (2025) using, for instance, an API for the matching descriptions or aggregating listings.

- Geocoding to fill the ZIP hole: use a geocoding API (for example, Google Maps Geocoding API) to complete the 30.4% of rows with missing zip_numeric, based on the results of the ablation study, since location plays a significant part in prediction.

- Hold-out based on time: switch from a random stratified train/test split to a time-based train/test split so that the test set is really prospective listings.

9. Conclusion

- This paper has shown that the well-designed multi-source pipeline involved in aggregating MICE imputation, state name normalization, geographic target encoding without leakage, VADER sentiment analysis NLP profiling, Bayesian hyperparameter optimization using Optuna, and Ridge Stacking performs sufficiently well in delivering an R^2 of 0.7869 on the test set and a mean absolute error of $62,473 on a real-dollar scale in a merged TX+NY dataset of 119,979 rows. This pipeline solves all the research problems stated in the research questions.

- Research question 1, which asks whether the ridge-stacked model performs better than any of the tuned base learners, is positively answered. The stacked ensemble model performs better because of the different errors from Random Forest which is the most important one, HGB which is the second most important), MLP is the third most important, and CatBoost (least important), which are consistent with theoretical prediction of Zhao and Wang (2023).

- This answer to RQ2 comes from a quantitative analysis: the three-way ablation experiment shows that the contribution to the decrease in RMSE for location encodings amounts to 0.3073, or 99.6% of the total contribution of the whole feature group, which is compared to just 0.0013 for the state-level NLP profile. The implication is not that the list description is inherently meaningless, but rather that this is a design limitation in the dataset because the descriptions are available only at the state level. Both the SHAP and permutation importance analyses confirm the same hierarchy independently using two different methodologies, reaching 5/5 overlap in top features throughtout all the three methods.

- RQ3 is answered with a qualified "yes" in that the HGB model provides a test AUC-ROC of 0.7707, which would be good enough for a practical listings which prioritizes application but would fall short for an accurate prediction of the individual listings. The four business recommendations is made on the basis of this research — including ZIP-code level benchmarking, investing in listing description quality, employing stacking ensembles, and using HGB to prioritize property portfolios — are all based explicitly on the output of the interpretability analysis and include concrete action thresholds (such as $62,473 listing price confidence interval; 0.25 probability of being the sold threshold) rather than being general advice.

- So the complete pipeline using these 4 datasets are executed by connecting seven colab notebooks and also engineering robustness.

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Appendix:

The complete code pipeline is present and as a version controlled source in the GitHub repository and the link is present on the title page.